

LAB SESSION 6

Programming Exercise:

Question 1: Create a class Rectangle with attributes length and width. Implement methods to calculate the area and perimeter of the rectangle.

```
#include <iostream>
using namespace std;

class Rectangle {
private:
    double length;
    double width;

public:
    // Constructor to initialize length and width
    Rectangle(double l, double w) {
        length = l;
        width = w;
    }

    // Method to calculate the area of the rectangle
    double calculateArea() {
        return length * width;
    }

    // Method to calculate the perimeter of the rectangle
    double calculatePerimeter() {
        return 2 * (length + width);
    }

    // Methods to set length and width (optional)
    void setLength(double l) {
        length = l;
    }

    void setWidth(double w) {
        width = w;
    }

    // Methods to get length and width (optional)
    double getLength() {
        return length;
    }

    double getWidth() {
        return width;
    }
};

int main() {
    // Create a rectangle object with length 5 and width 3
    Rectangle rect(5, 3);

    // Display area and perimeter
    cout << "Area of the rectangle: " << rect.calculateArea() << endl;
    cout << "Perimeter of the rectangle: " << rect.calculatePerimeter() <<
endl;
    return 0;
}
```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Area of the rectangle: 15
Perimeter of the rectangle: 16
PS C:\C++ PROGRAMS>
```

Question2: Create a class Circle with attribute radius. Implement methods to calculate the area and circumference of the circle.

```
#include <iostream>
#include <cmath> // For M_PI constant
using namespace std;

class Circle {
private:
    double radius;

public:
    // Constructor to initialize radius
    Circle(double r) {
        radius = r;
    }

    // Method to calculate the area of the circle
    double calculateArea() {
        return M_PI * radius * radius; // Area =  $\pi * r^2$ 
    }

    // Method to calculate the circumference of the circle
    double calculateCircumference() {
        return 2 * M_PI * radius; // Circumference =  $2 * \pi * r$ 
    }

    // Methods to set and get radius (optional)
    void setRadius(double r) {
        radius = r;
    }

    double getRadius() {
        return radius;
    }
};

int main() {
    // Create a Circle object with radius 5
    Circle circle(5);

    // Display area and circumference
    cout << "Area of the circle: " << circle.calculateArea() << endl;
    cout << "Circumference of the circle: " << circle.calculateCircumference() << endl;
    return 0;
}
```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Area of the circle: 78.5398
Circumference of the circle: 31.4159
PS C:\C++ PROGRAMS>
```

Question3: Create a class Employee with attributes name and salary. Implement a method to display the details of the employee.

```
#include <iostream>
#include <string> // For string handling
using namespace std;

class Employee {
private:
    string name;
    double salary;

public:
    // Constructor to initialize name and salary
    Employee(string n, double s) {
        name = n;
        salary = s;
    }

    // Method to display the details of the employee
    void displayDetails() {
        cout << "Employee Name: " << name << endl;
        cout << "Employee Salary: " << salary << endl;
    }

    // Methods to set and get name and salary (optional)
    void setName(string n) {
        name = n;
    }

    void setSalary(double s) {
        salary = s;
    }

    string getName() {
        return name;
    }

    double getSalary() {
        return salary;
    }
};

int main() {
    // Create an Employee object with name "John Doe" and salary 50000
    Employee emp("John Doe", 50000);

    // Display the details of the employee
    emp.displayDetails();

    return 0;
}
```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Employee Name: John Doe
Employee Salary: 50000
PS C:\C++ PROGRAMS>
```

Question4: Create a class BankAccount with attributes accountNumber, accountHolder, and balance. Implement methods to deposit and withdraw money from the account.

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Account Number: 101
Account Holder: John Doe
Balance: $1000
Deposited: $500
Withdrew: $300
Account Number: 101
Account Holder: John Doe
Balance: $1200
Insufficient balance!
PS C:\C++ PROGRAMS>
```

```

#include <iostream>
#include <string>
using namespace std;

class BankAccount {
private:
    int accountNumber;
    string accountHolder;
    double balance;

public:
    // Constructor to initialize the bank account details
    BankAccount(int accNumber, string holder, double initialBalance) {
        accountNumber = accNumber;
        accountHolder = holder;
        balance = initialBalance;
    }

    // Method to deposit money into the account
    void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            cout << "Deposited: $" << amount << endl;
        } else {
            cout << "Deposit amount must be positive!" << endl;
        }
    }

    // Method to withdraw money from the account
    void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            cout << "Withdrew: $" << amount << endl;
        } else if (amount > balance) {
            cout << "Insufficient balance!" << endl;
        } else {
            cout << "Withdrawal amount must be positive!" << endl;
        }
    }

    // Method to display account details
    void displayAccountDetails() {
        cout << "Account Number: " << accountNumber << endl;
        cout << "Account Holder: " << accountHolder << endl;
        cout << "Balance: $" << balance << endl;
    }
};

int main() {
    // Create a BankAccount object with account number 101, holder "John Doe", and an initial balance of $1000
    BankAccount account(101, "John Doe", 1000);

    // Display the initial account details
    account.displayAccountDetails();

    // Deposit money into the account
    account.deposit(500);

    // Withdraw money from the account
    account.withdraw(300);

    // Display updated account details
    account.displayAccountDetails();

    // Try to withdraw more money than the available balance
    account.withdraw(1500);

    return 0;
}

```

Question5: Create a class Car with attributes brand, model, and year. Implement a method to display the details of the car.

```
#include <iostream>
#include <string> // For string handling
using namespace std;

class Car {
private:
    string brand;
    string model;
    int year;

public:
    // Constructor to initialize car details
    Car(string b, string m, int y) {
        brand = b;
        model = m;
        year = y;
    }

    // Method to display car details
    void displayDetails() {
        cout << "Car Brand: " << brand << endl;
        cout << "Car Model: " << model << endl;
        cout << "Car Year: " << year << endl;
    }
};

int main() {
    // Create a Car object with brand "Toyota", model "Corolla", and year 2020
    Car car("Toyota", "Corolla", 2020);

    // Display the details of the car
    car.displayDetails();

    return 0;
}
```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Car Brand: Toyota
Car Model: Corolla
Car Year: 2020
PS C:\C++ PROGRAMS>
```

Question6: Create a class Fraction with attributes numerator and denominator. Implement a method to simplify the fraction.

```
#include <iostream>
using namespace std;

class Fraction {
private:
    int numerator;
    int denominator;

    // Helper function to find the Greatest Common Divisor (GCD) using Euclid's
    algorithm
    int gcd(int a, int b) {
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }
        return a;
    }

public:
    // Constructor to initialize the fraction with numerator and denominator
    Fraction(int num, int denom) {
        if (denom == 0) {
            cout << "Denominator cannot be zero!" << endl;
            numerator = 0;
            denominator = 1; // Set to a default value
        } else {
            numerator = num;
            denominator = denom;
        }
    }

    // Method to simplify the fraction
    void simplify() {
        int commonDivisor = gcd(numerator, denominator);
        numerator /= commonDivisor;
        denominator /= commonDivisor;
    }

    // Method to display the fraction
    void display() {
        cout << "Fraction: " << numerator << "/" << denominator << endl;
    }
};

int main() {
    // Create a Fraction object with numerator 8 and denominator 12
    Fraction frac(8, 12);

    // Display the original fraction
    cout << "Original ";
    frac.display();

    // Simplify the fraction
    frac.simplify();

    // Display the simplified fraction
    cout << "Simplified ";
    frac.display();

    return 0;
}
```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($  
?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }  
Original Fraction: 8/12  
Simplified Fraction: 2/3  
PS C:\C++ PROGRAMS>
```